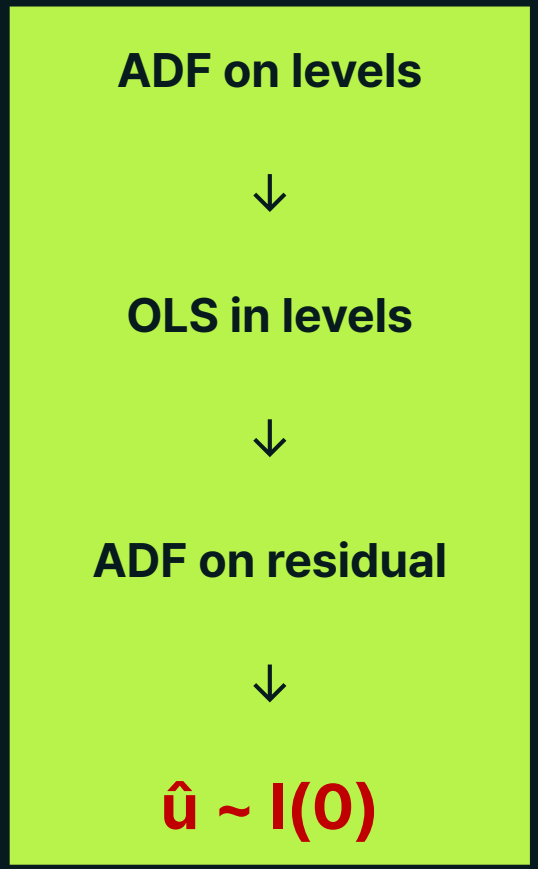
Two nonstationary series share one long-run path

The Engle–Granger two-step cointegration procedure



Two regressions and two unit-root tests settle it

01	Why it matters	**Nonstationarity** Spurious regression · the I(d) ladder
02	Step 0	**Order of integration** ADF on each level · both I(1) · never I(2)
03	Steps 1 and 2	**Long-run OLS, then the residual** Levels regression · residual û · residual ADF
04	Cautions and use	**Critical values, then a trade** MacKinnon values · normalization · pairs trade

Cointegration is a claim about the residual, not about the fit

The whole procedure fits on one page



DECISION RULE

Reject H_0 → **COINTEGRATED**

Step 0 and Step 2 are unit-root tests; Step 1 is the regression that links them.

A stationary series comes back; a nonstationary one does not

STATIONARY · I(0)

It comes back

- Shocks fade away
- Fixed mean, fixed variance
- t and F tests are valid

White noise, AR(1) with $|\varphi| < 1$

NONSTATIONARY · I(1)

The level wanders

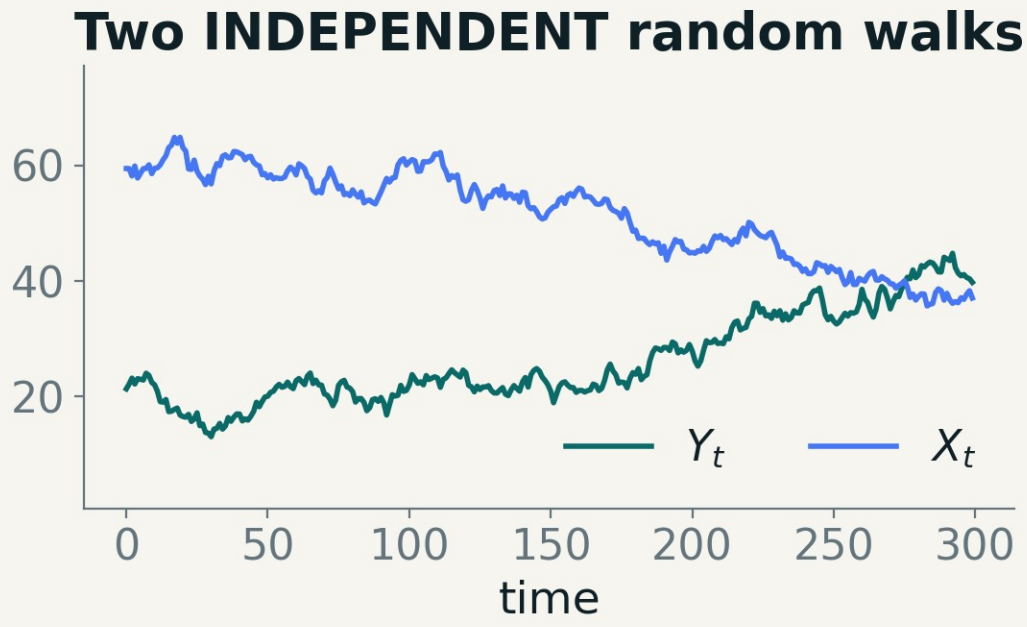
- Shocks are permanent
- No mean to return to
- OLS inference breaks down

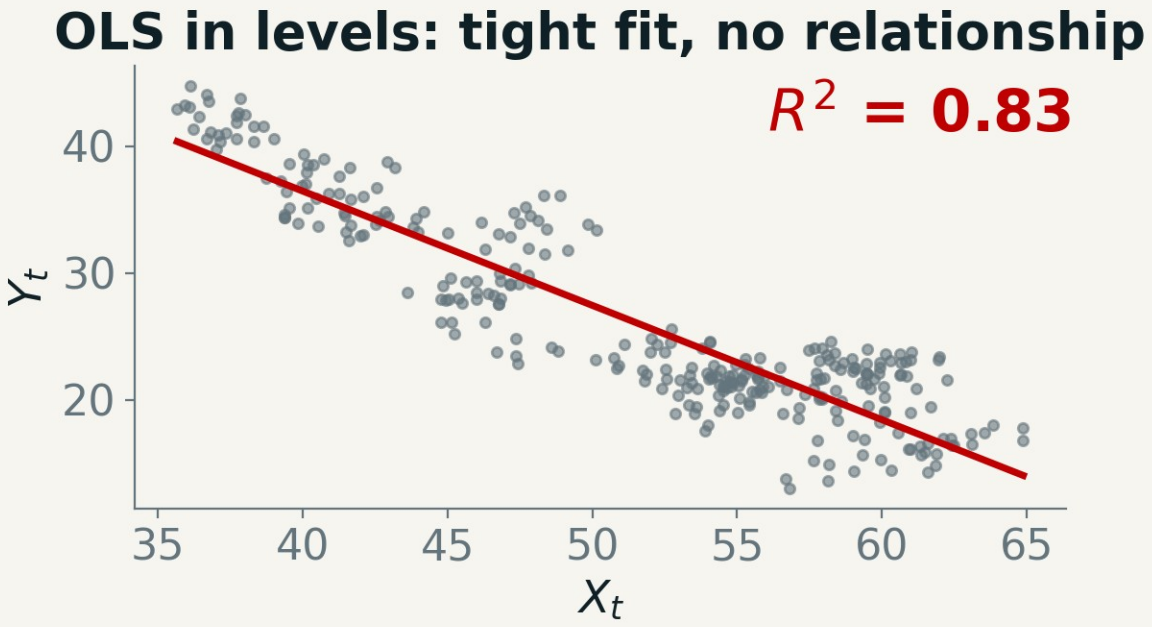
Random walk, most asset prices

Differencing restores stationarity but discards the level relationship.

Cointegration keeps the levels and still gives valid inference.

Regressing two random walks invents a relationship





A tight fit and a large t-statistic prove nothing while the residual wanders.

※ Simulated: two independent random walks, T = 300 — the residual here is I(1)

The order of integration counts the differences you need

$$\Delta^d Y_t \sim I(0) \iff Y_t \sim I(d)$$

I(0)

Already stationary

Levels revert to a mean

I(1)

One difference

The random-walk family

I(2)

Two differences

Breaks this recipe

Engle–Granger assumes both variables sit on the same rung: I(1).

Cointegration means one linear combination is stationary

Definition

$$Y_t \sim I(1), \quad X_t \sim I(1) \qquad \text{but} \qquad Y_t - \beta X_t \sim I(0)$$

WHAT IT MEANS

A long-run tether

Deviations are temporary

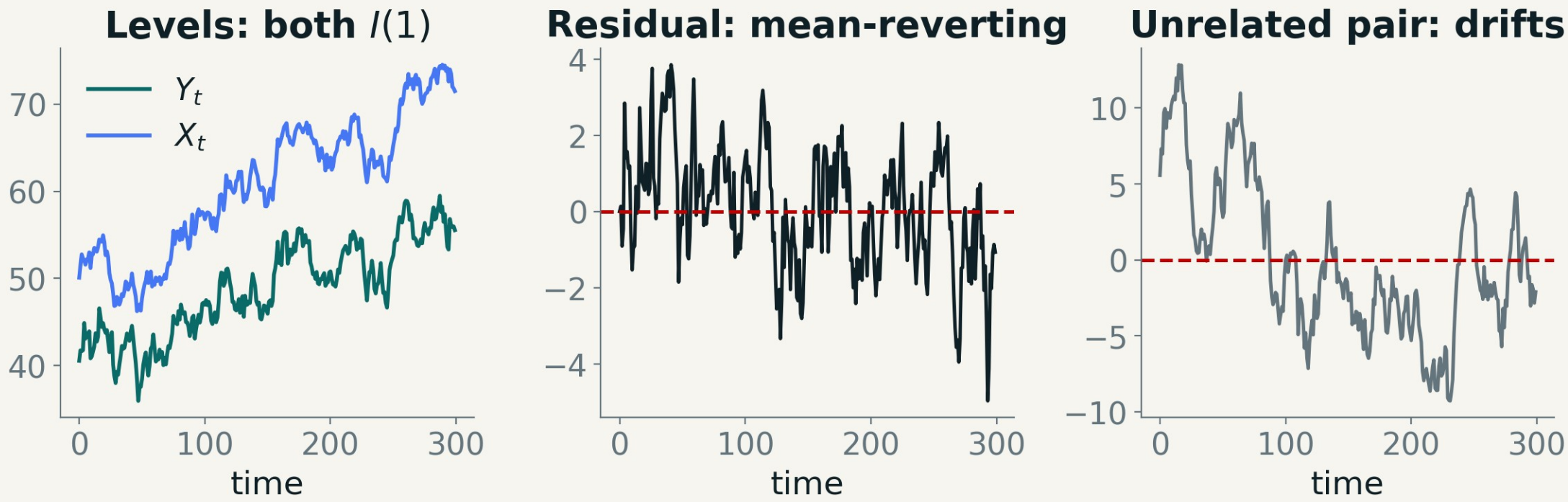
WHY WE CARE

β is the hedge ratio

The basis of pairs trading

Individually I(1), jointly I(0) — that is the entire claim.

Cointegration looks like a spread that returns to its mean



※ Simulated: Y and X are both I(1); only the left pair leaves a stationary residual

01

Step 0 — Establish the order of integration

Before any cointegration test, pin down I(d) for every variable

ADF asks whether the level series carries a unit root

ADF regression, run on the level of each variable

$$\Delta Y_t = \mu + \gamma Y_{t-1} + \sum_{i=1}^{p} \phi_i \, \Delta Y_{t-i} + \varepsilon_t$$

Hypotheses

$$H_0 : \ \gamma = 0 \ \ (\text{unit root}) \qquad H_1 : \ \gamma < 0 \ \ (\text{stationary})$$

Fail to reject in levels, reject in first differences → the series is I(1).

※ Include a trend term only if the level series shows a clear deterministic trend

Both variables must be I(1), and neither may be I(2)

Levels are nonstationary

$$Y_t \sim I(1), \qquad X_t \sim I(1)$$

First differences are stationary

$$\Delta Y_t \sim I(0), \qquad \Delta X_t \sim I(0)$$

Mixed orders void the procedure: an I(2) variable cannot be paired with an I(1) one.

※ If one variable is I(0), no cointegration question arises — plain OLS already applies

02

Step 1 — Estimate the long-run relationship

One OLS regression in levels, and the residual it leaves behind

OLS in levels delivers the long-run coefficient

Cointegrating regression

$$Y_t = \alpha + \beta X_t + u_t$$

INTERCEPT

Long-run level α

SLOPE

Hedge ratio β

RESIDUAL

Disequilibrium u

β is the long-run equilibrium coefficient — the hedge ratio of the pair.

The residual measures today's departure from equilibrium

Equilibrium error

$$\hat{u}_t = Y_t - \hat{\alpha} - \hat{\beta} X_t$$

$\hat{\beta}$ is super-consistent: $\hat{\beta} - \beta = O_p(T^{-1})$

Cointegration is decided by whether û is stationary, not by how good the fit is.

※ Fast convergence of β does not make its standard t-statistic usable

03

Step 2 — Test the residual for a unit root

A residual-based ADF, read against cointegration critical values

A residual-based ADF decides the question

Residual ADF regression

$$\Delta \hat{u}_t = \rho\, \hat{u}_{t-1} + \sum_{i=1}^{p} \gamma_i\, \Delta \hat{u}_{t-i} + \varepsilon_t$$

LAG ORDER p

AIC or BIC

Whiten the residual

NO CONSTANT

û has zero mean

By OLS construction

ONE-SIDED

Left tail only

Reject when ρ < 0

The lag terms exist only to clean up serial correlation in Δû.

The null hypothesis is no cointegration

H_0 — the residual has a unit root

$$H_0: \ \rho = 0 \quad \Rightarrow \quad \hat{u}_t \sim I(1) \ \ (\text{no cointegration})$$

H_1 — the residual is stationary

$$H_1: \ \rho < 0 \quad \Rightarrow \quad \hat{u}_t \sim I(0) \ \ (\text{cointegration})$$

Rejection is the evidence for cointegration.

Failing to reject is weak evidence, not proof of no relationship.

Rejecting the null makes the pair cointegrated

Stationary long-run combination

$$Y_t - \hat{\alpha} - \hat{\beta} X_t \sim I(0)$$

The residual û then becomes the error-correction term of the ECM

From here on, the spread — not the price — is the object you model and trade.

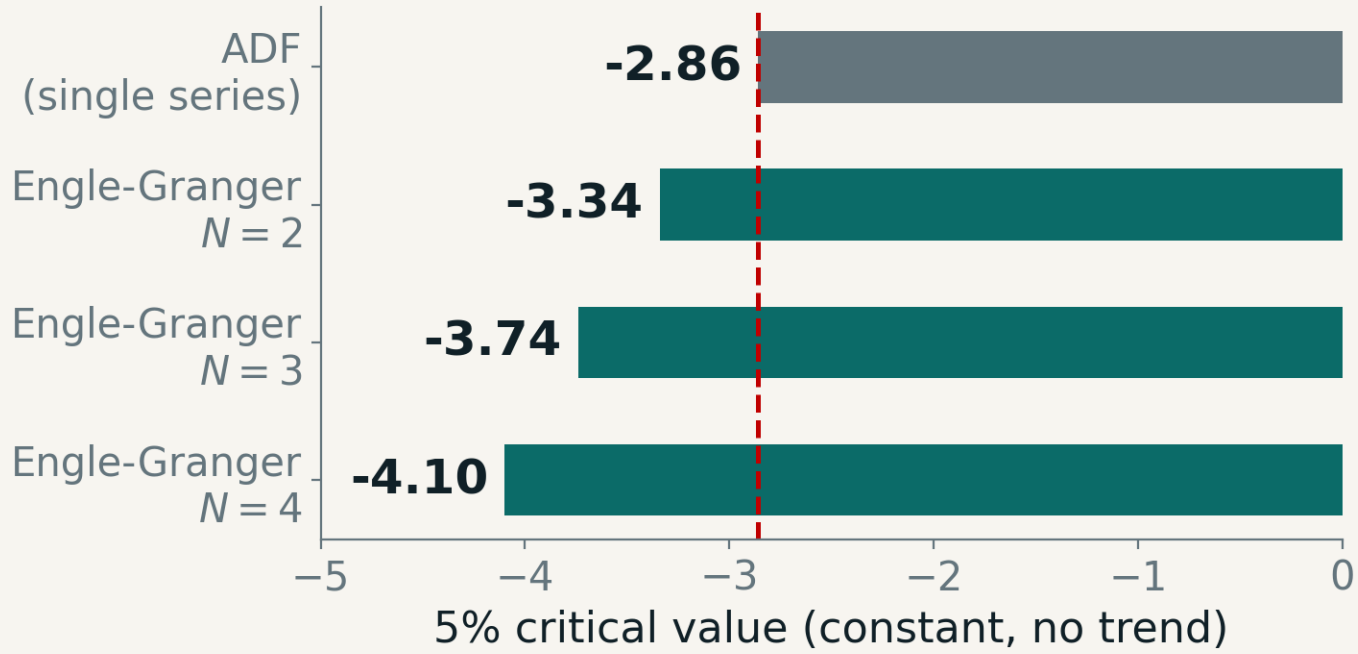
Do not read this test off an ordinary ADF table

CRITICAL VALUES

Shifted left

- û is estimated
- OLS flattens it
- MacKinnon tables

ADF tables over-reject



The more variables in the regression, the further left the rejection bar moves.

※ 5% values, constant and no trend; N counts the variables in the levels regression

In practice, have Claude Code run the cointegration routine

PROMPT · Claude Code

In the coint-lab folder, do the following.

1. Simulate a pair with numpy default_rng(5), T = 400, and save it as pair.csv: X = 50 + random walk (drift 0.05, sd 1), u = AR(1) (φ = 0.75, sd 1), Y = 3 + 1.5X + u.
2. Run the Engle–Granger test with statsmodels coint (trend="c"), for Y on X and for X on Y.
3. Save the t-statistic, p-value and 5% critical value of both directions to coint_result.csv.
4. Do not run adfuller on the residual yourself. In R, use ca.po from urca or po.test from tseries.

CHECK ① Compare against the right table

$$t_{\hat{\rho}} < \mathrm{CV}^{\mathrm{EG}}_{N,5\%}\ \ (\mathrm{MacKinnon}) \quad \neq \quad t_{\hat{\rho}} < -2.86\ \ (\mathrm{ADF})$$

CHECK ②

Normalization

Y on X ≠ X on Y

Hand-rolling an ADF on the residual is where the critical values go wrong.

With three or more variables, move to the Johansen system test.

A pairs trade is the residual, traded back to zero

Spread and its standardized score

$$\hat{u}_t = Y_t - \hat{\alpha} - \hat{\beta} X_t, \qquad z_t = \frac{\hat{u}_t}{\hat{\sigma}_{\hat{u}}}$$

ENTRY

z above +2

Short Y, long β of X

EXIT

z back to 0

Equilibrium restored

STOP

z beyond ±3

The relation may be gone

The tradable object is û — the hedge ratio β comes straight out of Step 1.

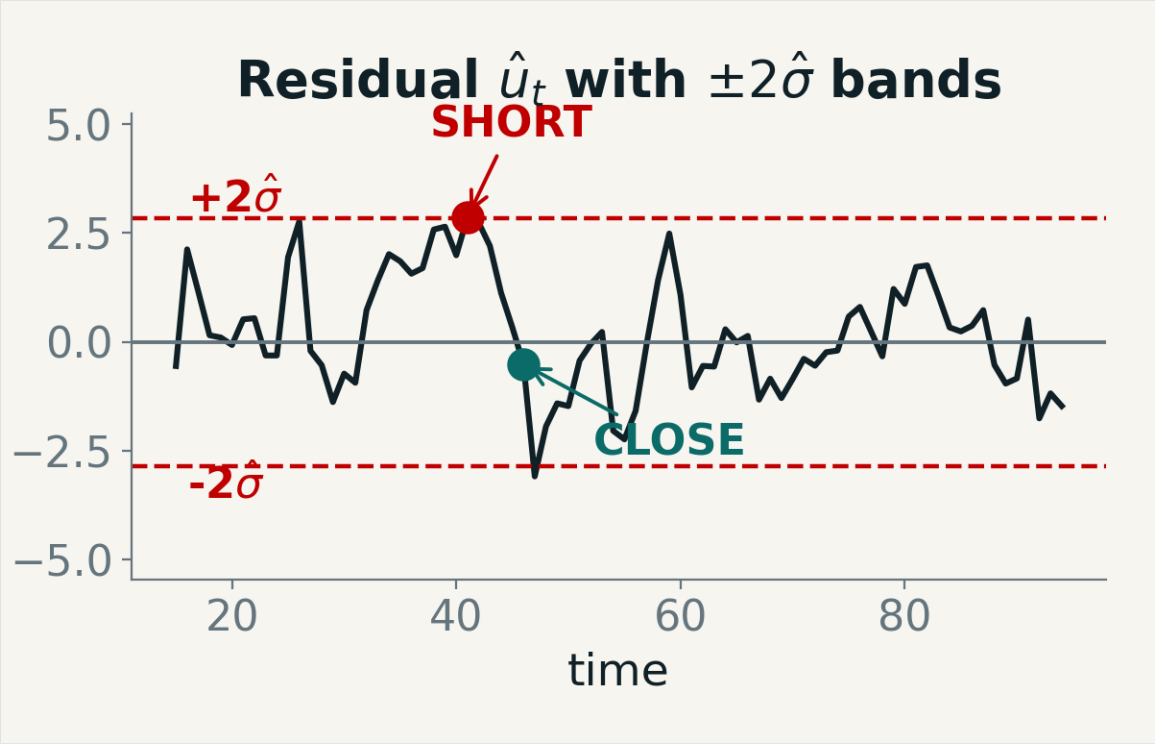
Example: short the spread at +2σ, close at the mean

THE TRADE

One round trip

- β = 0.67, σ̂ = 1.42
- Enter at z = +2.0
- Close at z = 0

Held for 5 periods



Mean reversion of û is what pays; the level of Y and X never has to revert.

※ Same simulated pair as above; entry and exit are the first band crossing after t = 40

The arithmetic of one round trip

hedge ratio β	residual σ̂	entry û (z = +2.01)	exit û (z = −0.36)
0.67	**1.42**	**+2.86**	**−0.51**

P&L on Q = 100 spread units

$$\Pi = \left(\hat{u}_{t_0} - \hat{u}_{t_1}\right) \times Q = \left(2.86 - (-0.51)\right) \times 100 \approx 337$$

Short 1 unit of Y against 0.67 units of X, and the gain is the fall in û.

※ Illustrative: gross of transaction costs, financing and slippage

Both series wander.

Their spread does not.

That is cointegration.

Engle–Granger:
two steps, one decision

※ Next: Johansen's system test, for more than one cointegrating vector